

Bug Report: YouTube Redirect from Search Queries

Date: February 17, 2026
Component: KomalSafetyScannerViewModel / Browser Navigation
Severity: **HIGH**

1. Bug Description

When a user types certain search words (e.g., 'funny videos', 'music', 'gaming') into the Komal browser's omnibox, the browser navigates directly to YouTube instead of displaying Google search results. This happens without the user clicking any YouTube link.

2. Root Cause Analysis

The issue originates from how the app handles server scan responses. When a user enters a search term, the app correctly converts it to a Google Safe Search URL:

```
https://www.google.com/search?q=funny+videos&safe=active
```

This URL is sent to the Komal server for content analysis. The server fetches the page, follows any redirects, and returns a UnifiedDecisionResponse or ScanResponse. Critically, the response includes a 'url' field that may differ from the original URL sent.

For search terms related to video content, Google's top results point to YouTube. The server may resolve the destination and return the YouTube URL in the response:

```
Server response: decision.url = 'https://www.youtube.com/results?search_query=funny+videos'
```

The app then navigates to decision.url instead of the original Google search URL, causing an unintended redirect to YouTube.

3. Affected Code Paths

Path A: Unified Decision (Primary)

File: KomalSafetyScannerViewModel.swift
Function: handleUnifiedAction(_:decision:)

```
// BEFORE (Bug): navigates to server-returned URL
case .allow:
    if let url = URL(string: decision.url) { // <-- server URL
        lastSafeURL = url
        currentURL = url // navigates to YouTube!

case .gate:
    if let url = URL(string: decision.url) { // <-- server URL
        pendingURL = url // stores YouTube URL for later
```

Path B: Legacy Scan (Fallback)

File: KomalSafetyScannerViewModel.swift

Function: handleAction(_:result:)

```
// BEFORE (Bug): same pattern with legacy response
case .allow:
    if let url = URL(string: result.url) { // <-- server URL
        currentURL = url // navigates to YouTube!

case .gate:
    if let url = URL(string: result.url) {
        pendingURL = url // stores YouTube URL
```

Path C: Fallback when no age action found

File: KomalSafetyScannerViewModel.swift

Function: processScanResult(normalizedURL:)

```
// BEFORE (Bug): fallback uses server URL
if let url = URL(string: result.url) {
    currentURL = url // navigates to YouTube!
```

4. Fix Applied

All three code paths were updated to use the original normalizedURL (the Google search URL that the app constructed) for navigation, instead of the server-returned URL. The server URL is still used for logging and analytics.

Fix A: handleUnifiedAction

```
// AFTER (Fixed): uses original URL for navigation
private func handleUnifiedAction(
    _ action: Action,
    decision: UnifiedDecisionResponse,
    originalURL: String          // <-- NEW parameter
) {
    case .allow:
        if let url = URL(string: originalURL) { // <-- original URL
            currentURL = url // stays on Google search
        }
    case .gate:
        if let url = URL(string: originalURL) {
            pendingURL = url // stores original URL
        }
}
```

Fix B: handleAction (legacy)

```
// AFTER (Fixed): same pattern for legacy path
private func handleAction(
    _ action: Action,
    result: ScanResponse,
    originalURL: String          // <-- NEW parameter
) {
    case .allow:
        if let url = URL(string: originalURL) {
            currentURL = url // stays on Google search
        }
    case .gate:
        if let url = URL(string: originalURL) {
            pendingURL = url
        }
}
```

Fix C: processScanResult fallback

```
// AFTER (Fixed): fallback uses normalizedURL
if let url = URL(string: normalizedURL) { // <-- original URL
    currentURL = url
}
```

5. Call Sites Updated

- processUnifiedWithAction() now passes normalizedURL to handleUnifiedAction()
- processScanResult() now passes normalizedURL to handleAction()

6. Additional Fixes (Previous Session)

Default/fallback redirect URLs were also changed from external platforms to Google:

- BrowserView.swift: Intervention dismissal redirect changed from khanacademy.org to google.com
- BrowserViewModel.swift: Default browser URL changed from khanacademy.org to google.com

7. Reproduction Steps (Before Fix)

- 1. Open Komal browser
- 2. Type 'funny videos' or 'music' in the omnibox
- 3. Press Enter/Submit
- 4. Observe: browser navigates directly to YouTube instead of Google search results

8. Expected Behavior (After Fix)

- Search terms always show Google search results first
- YouTube is accessible if user clicks a YouTube link in search results
- YouTube is not blocked (no intervention popup)
- App never automatically redirects to YouTube or any external platform

9. Build Verification

All changes compile successfully:

```
$ xcodebuild build -scheme Komalios \  
  -destination 'generic/platform=iOS' \  
  CODE_SIGNING_ALLOWED=NO
```

```
** BUILD SUCCEEDED **
```

10. Files Modified

```
Sources/Komalios/PostAuth/BrowserView/ViewModel/KomalSafetyScannerViewModel.swift  
Sources/Komalios/PostAuth/BrowserView/View/BrowserView.swift  
Sources/Komalios/PostAuth/BrowserView/ViewModel/BrowserViewModel.swift
```